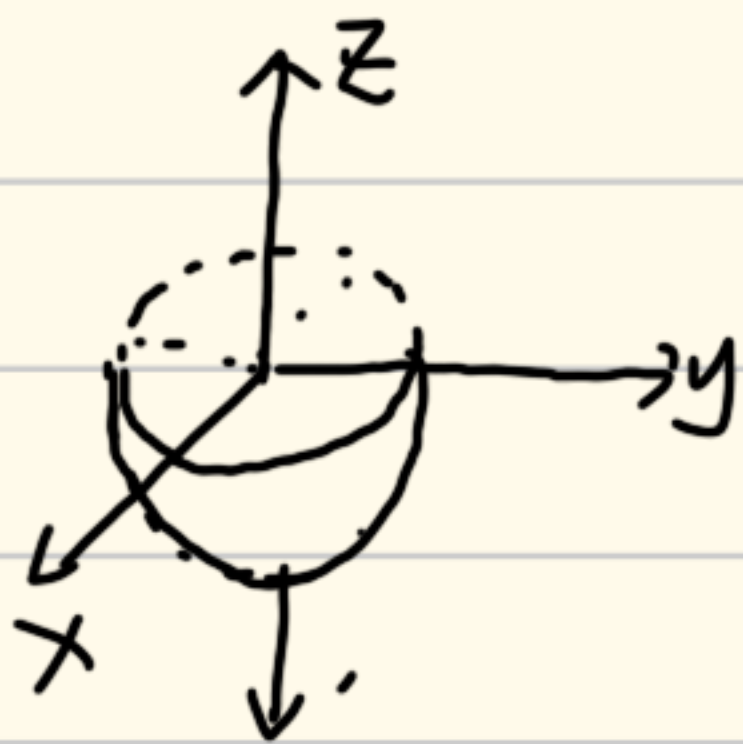


③ 求 $\iint_S x^2 y^2 z dx dy$, S 是球面 $x^2 + y^2 + z^2 = R^2$ 下半部的下侧.



$$\begin{aligned}
 z &= -\sqrt{R^2 - x^2 - y^2} \\
 \iint_S x^2 y^2 \sqrt{R^2 - x^2 - y^2} dx dy \\
 &= \int_0^{2\pi} d\theta \int_0^R r^5 \cos^2 \theta \sin^2 \theta \sqrt{R^2 - r^2} dr \\
 &= \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} R^5 \sin^2 \theta \cos^2 \theta \sin^5 \varphi \cdot R^2 \sin^2 \varphi d\varphi \\
 &= R^7 \int_0^{2\pi} \sin^2 \theta \cos^2 \theta d\theta \int_0^{\frac{\pi}{2}} \sin^5 \varphi \cos^2 \varphi d\varphi \\
 &= R^7 \cdot \frac{2}{\pi} \cdot \frac{8}{105} \\
 &= \frac{2}{105} \pi R^7
 \end{aligned}$$

⑤ 求 $\iint_S xy^2 dy dz + yz^2 dz dx + zx^2 dx dy$, 其中 S 为椭球面 $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ 的外侧.

$$\begin{aligned}
 5. \quad S_1: z &= c \sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}} \\
 z_x &= \frac{-\frac{2x}{a^2}}{2c \sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}}} = \frac{-x}{a^2 \sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}}} = \frac{-c^2 x}{a^2 z} \\
 z_y &= \frac{-c^2 y}{b^2 z}
 \end{aligned}$$

$$\begin{aligned}
 S_1 \text{ 外侧为正} \quad \iint_S (xy^2 \cdot \frac{c^2 x}{a^2 z} + yz^2 \cdot \frac{c^2 y}{b^2 z} + zx^2) dx dy \\
 \text{令 } x = ar \cos \theta, \quad y = br \sin \theta, \quad z = c \sqrt{1 - r^2} \\
 |J| = ab \cdot r
 \end{aligned}$$

$$\text{原式} = \iint_S \left(\frac{ca^2b^2r^4\sin^2\theta\cos^2\theta}{a^2\sqrt{1-r^2}} + \frac{cb^2r^2\sin^2\theta\cos^2\theta}{b^2} + c\sqrt{1-r^2}\cdot a^2r^2\cos^2\theta \right) ab r \cdot d\theta dr$$

$$= ab \int_0^{2\pi} d\theta \cdot \int_0^1 \left(cb^2\sin^2\theta\cos^2\theta \frac{r^5}{\sqrt{1-r^2}} + \sin^2\theta \cdot c^3r^3\sqrt{1-r^2} + a^2c\cos^2\theta \cdot r^3\sqrt{1-r^2} \right) dr$$

$$= ab \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} \left(cb^2\sin^2\theta\cos^2\theta \sin^5\varphi + \sin^2\theta \cdot c^3\sin^3\varphi\cos^2\varphi + a^2c\cos^2\theta \cdot \sin^3\varphi\cos^2\varphi \right) d\varphi$$

$$= ab \int_0^{2\pi} \left(\frac{8}{15} \cdot cb^2 \cdot \sin^2\theta\cos^2\theta + \frac{2}{15} \sin^2\theta \cdot c^3 + a^2c \cdot \cos^2\theta \right) d\theta$$

$$= ab \int_0^{2\pi} \left(\frac{2}{15} b^2c + \frac{2}{15} c^3 + a^2c \right) d\theta$$

$$= \frac{2}{15} abc (a^2 + b^2 + c^2) 2\pi$$

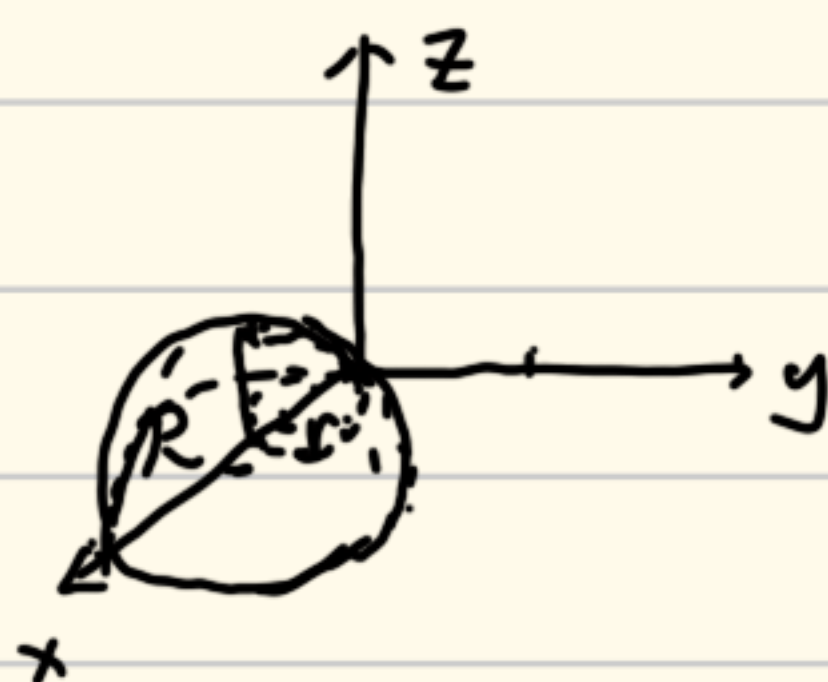
$$S_2: z = -c\sqrt{1-\frac{x^2}{a^2}-\frac{y^2}{b^2}}, \quad z_x = \frac{c^2x}{a^2z}, \quad z_y = \frac{c^2y}{b^2z}$$

$$S_2 \text{ 外侧为负向} \quad \text{原式} = \iint_S \left(xy^2 \frac{c^2x}{a^2z} + yz^2 \frac{c^2y}{b^2z} + zx^2 \right) dx dy = \frac{2}{15} abc (a^2 + b^2 + c^2) 2\pi$$

$$\therefore \text{总} = \frac{4}{15} abc (a^2 + b^2 + c^2) 2\pi$$

⑥ 求 $\iint_S (y-z)dydz + (z-x)dzdx + (x-y)dxdy$, 其中 S 是上半球面

$x^2 + y^2 + z^2 = 2Rx (z \geq 0)$ 被柱面 $x^2 + y^2 = 2rx (0 < r < R)$ 所截部分的 上侧 (提示: 化为二重积分后利用对称性化简.)



$$(x-R)^2 + y^2 + z^2 = R^2$$

$$S_1: z = \sqrt{R^2 - y^2 - (x-R)^2} = \sqrt{2Rx - x^2 - y^2}$$

$$z_x = \frac{2R-2x}{2\sqrt{2Rx-x^2-y^2}} = \frac{R-x}{z}$$

$$z_y = \frac{-2y}{2\sqrt{2Rx-x^2-y^2}} = \frac{-y}{\sqrt{2Rx-x^2-y^2}} = -\frac{y}{z}$$

上侧为正向

$$\text{原式} = \iint_S \left(\frac{xR}{z} (y-z) + \frac{y}{z} (z-x) + x-y \right) dx dy$$

$$= \iint_S \left(\frac{xR}{z} y - x + R + y - \frac{xy}{z} + x - y \right) dx dy$$

$$= \iint_S R \left(1 - \frac{y}{z} \right) dx dy$$

积分区域关于 xOz 面对称, 而 $\frac{y}{z}$ 为关于 y 的奇函数

$$\text{则 } \iint_R (-\frac{y}{z}) dx dy = 0$$

$$\therefore \text{原式} = R \iint_S dx dy = R \cdot \pi \cdot r^2 = \pi r^2 \cdot R$$

(10.) 求 $\iint_S x dy dz + y dz dx + z dx dy$, 其中 S 为螺旋面 $x = u \cos v, y = u \sin v, z = cv$

($0 \leq a \leq u \leq b, 0 \leq v \leq 2\pi$) 的上侧. (提示: 先化为第一型曲面积分.)

$$A = \frac{D(y, z)}{D(u, v)} = \begin{vmatrix} y_u & y_v \\ z_u & z_v \end{vmatrix} = \begin{vmatrix} \sin v & u \cos v \\ 0 & c \end{vmatrix} = c \sin v.$$

$$B = \frac{D(z, x)}{D(u, v)} = \begin{vmatrix} 0 & c \\ \cos v & -u \sin v \end{vmatrix} = -c \cos v.$$

$$C = \frac{D(x, y)}{D(u, v)} = \begin{vmatrix} \cos v & -u \sin v \\ \sin v & u \cos v \end{vmatrix} = u$$

$$\iint_S (u \cos v \cdot c \sin v - c \cos v \cdot u \sin v + u \cdot cv) du dv.$$

$$= c \iint_S uv du dv$$

$$= c \cdot \int_a^b u du \int_0^{2\pi} v dv$$

$$= \frac{1}{2} c \cdot (b^2 - a^2) \cdot \frac{1}{2} \cdot 4\pi^2.$$

$$= c(b^2 - a^2)\pi^2$$

1. 利用高斯公式, 求第二型曲面积分

$$\iint_S x^2 dy dz + y^2 dz dx + z^2 dx dy,$$

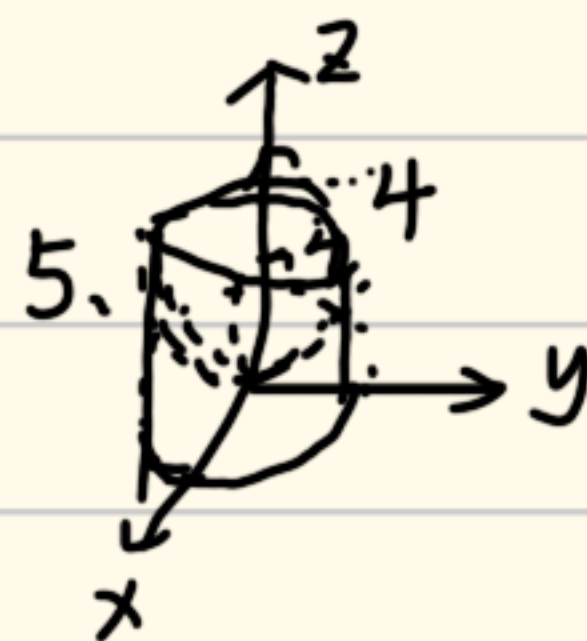
其中 S 分别为

(1) 圆柱面 $x^2 + y^2 = a^2$ ($0 \leq z \leq h$) 的外侧;

(2) 圆锥体 $\frac{x^2}{a^2} + \frac{y^2}{a^2} - \frac{z^2}{b^2} \leq 0$ ($0 \leq z \leq b$) 的全表面的外侧.

1. (2) 由高斯公式, 原式 =

$$\begin{aligned} & \iiint_V (2x + 2y + 2z) dV, \quad \begin{cases} x = r \cos \theta, y = r \sin \theta, z = z, \\ |J| = r. \end{cases} \\ & \quad \frac{1}{a^2} r^2 \leq \frac{z^2}{b^2}, \quad 0 \leq r \leq \frac{az}{b} \\ & 2 \int_0^{2\pi} d\theta \int_0^b dz \int_0^{\frac{az}{b}} (r^2 (\sin \theta + \cos \theta) + r \cdot z) dr \\ & = 2 \int_0^{2\pi} d\theta \int_0^b \left(\frac{1}{3} \cdot \frac{a^3 z^3}{b^3} (\sin \theta + \cos \theta) + \frac{a^2}{2b^2} z^3 \right) dz \\ & = 2 \int_0^{2\pi} \left[\frac{1}{12} a^3 b (\sin \theta + \cos \theta) + \frac{1}{8} a^2 b^2 \right] d\theta \\ & = \frac{2}{2} a^2 b^2 \end{aligned}$$



原式 = $\iiint_V x - 1 dV$, 由于图形关于 yz 面对称,

$$\text{则 } \iiint_V x dV = 0$$

$$\begin{aligned} \text{原式} &= - \iiint_V dV = - \int_0^{2\pi} d\theta \int_0^2 dr \int_0^{r^2} r \cdot dh \\ &= - 2\pi \cdot \int_0^2 r^3 \cdot dr \\ &= - 8\pi \end{aligned}$$

5. 求 $\oint_{S^+} p dy dz + q y dz dx - z dx dy$, 其中 S^+ 是圆柱 $x^2 + y^2 \leq 4$ 被 $z = 0$ 及 $z = x^2 + y^2$ 所截部分立体之 表面外侧.

6. 求 $\iint_{S^+} \mathbf{F} \cdot \mathbf{n} dS$, 其中 $\mathbf{F} = (x^4, y^4, z^4)$, S^+ 是锥面 $x^2 + y^2 = z^2$ 在 $0 \leq z \leq h$ 部分的外侧.

6. 补充平面 $S_0: x^2 + y^2 = h^2, z = h$.

则 $\iint_{S^+ + S_0^+} \vec{F} \cdot \vec{n} \cdot dS = \iiint_V (4x^3 + 4y^3 + 4z^3) dV$, 由对称性原式 =
 $4 \iiint_V z^3 dV = 4 \int_0^{2\pi} d\theta \int_0^h dr \int_0^r r z^3 dz$
 $= 8\pi \cdot \int_0^h \frac{1}{4} r^5 dr$
 $= 8\pi \cdot \frac{1}{24} h^6 = \frac{1}{3} \pi h^6$

而 $\iint_{S^+} \vec{F} \cdot \vec{n} \cdot dS = \iint_{S_0^+} h^4 dS = \pi h^6$

则 $\iint_{S^+} \vec{F} \cdot \vec{n} \cdot dS = -\frac{2}{3} \pi h^6$

9. 设向量函数 $F(x, y, z) = (\cos z, \sin z, 1)$, S 为封闭的简单曲面, 证明:

$$\oiint_{S^+} \cos(\vec{F}, \vec{n}) dS = 0,$$

其中 $\vec{n}$ 为曲面 S 的外法线方向的单位向量.

由高斯公式

$$\iint_{S^+} \frac{\vec{F} \cdot \vec{n}}{|\vec{F}| \cdot |\vec{n}|} \cdot dS = \frac{1}{|\vec{F}|} \cdot \iint_{S^+} (\cos z dy dz + \sin z dz dx + dx dy) = \frac{1}{\sqrt{2}} \iiint_V 0 dV = 0$$

证毕

11. 设 $u(x, y, z)$ 是三维调和函数, 即 u 有连续二阶偏导数, 且满足 $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} +$

$\frac{\partial^2 u}{\partial z^2} = 0$. 又设 S 为一光滑闭曲面, S 所围区域为 Ω . 证明:

(1) $\oiint_{S^+} u \frac{\partial u}{\partial \vec{n}} dS = \iiint_{\Omega} (u_x^2 + u_y^2 + u_z^2) dV$, 其中 $\frac{\partial u}{\partial \vec{n}}$ 为 u 沿 S 的外法线方向的方向导数;

(2) 若 $u(x, y, z)$ 在 S 上恒为零, 则 $u(x, y, z)$ 在区域 Ω 上也恒为零.

(1) 设 S 上某点正法方向的方向余弦为 $(\cos \alpha, \cos \beta, \cos \gamma)$

则 $\iint_{S^+} (u \cdot \frac{\partial u}{\partial x} \cos \alpha + u \cdot \frac{\partial u}{\partial y} \cos \beta + u \cdot \frac{\partial u}{\partial z} \cos \gamma) \cdot dS$
 $= \iint_{S^+} u \cdot \frac{\partial u}{\partial x} dy dz + u \cdot \frac{\partial u}{\partial y} dz dx + u \cdot \frac{\partial u}{\partial z} dx dy$

$$= \iiint_V u_x^2 + u_y^2 + u_z^2 + u \cdot \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) dV$$

$$= \iiint_V (u_x^2 + u_y^2 + u_z^2) dV$$

(2) 若 $u(x, y, z) \equiv 0, \forall (x, y, z) \in S$, 则有 $\iint_{S^+} u \cdot \frac{\partial u}{\partial n} ds = 0$

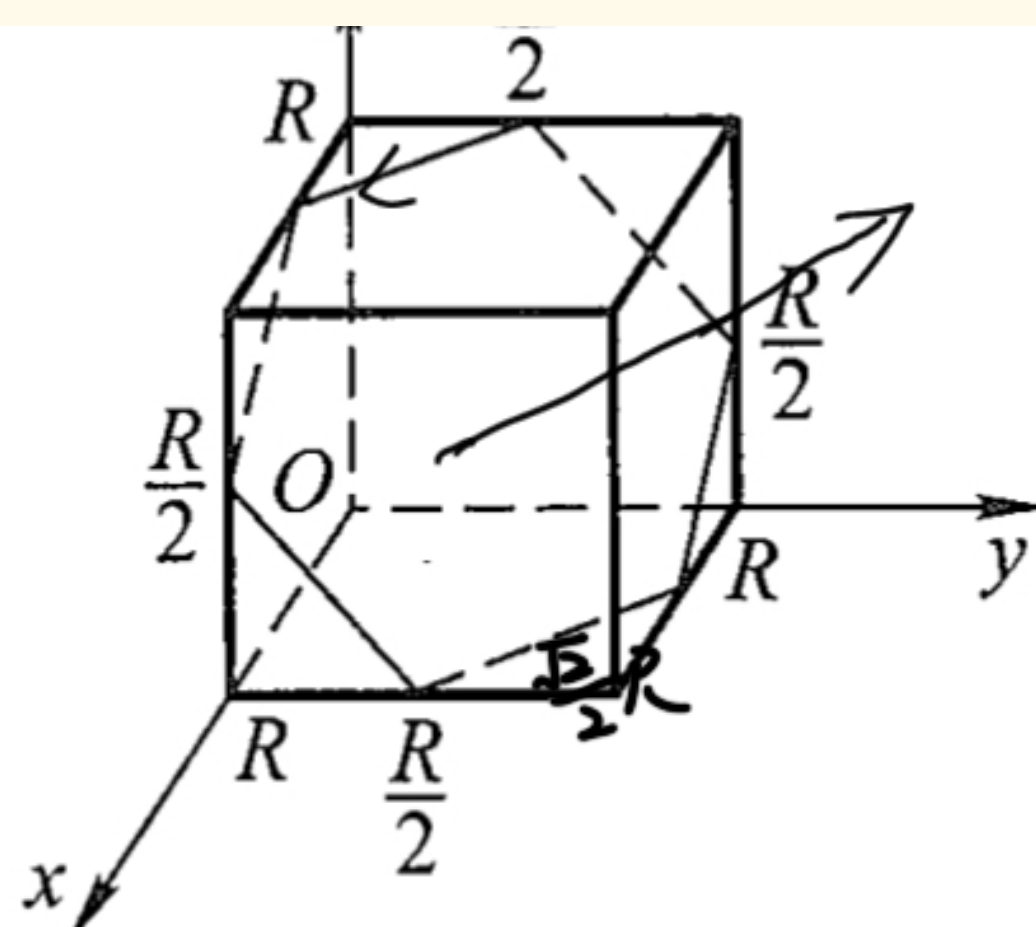
$$\text{则 } \iiint_V (u_x^2 + u_y^2 + u_z^2) dV = 0$$

由于各项非负, 则有 $u_x = u_y = u_z = 0$

则有 $u(x, y, z)$ 在 V 上必为常函数

又 $\because S \subset \partial V$, 则有 $u(x, y, z) \equiv 0, \forall (x, y, z) \in \partial V$.

14. 求 $\oint_{L^+} (y^2 - z^2 + x^2) dx + (z^2 - x^2 + y^2) dy + (x^2 - y^2 + z^2) dz$, 其中 L 是平面 $x + y + z = \frac{3}{2}R$ 与立方体 $\{(x, y, z) \mid 0 \leq x \leq R, 0 \leq y \leq R, 0 \leq z \leq R\}$ 的交线, 从 x 轴正向看去, L^+ 沿逆时针方向 (见图 8.50).



方向余弦为 $\frac{1}{\sqrt{3}}(1, 1, 1)$

$$\text{原式} = \iint_S \begin{vmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ \frac{\partial R}{\partial x} & \frac{\partial R}{\partial y} & \frac{\partial R}{\partial z} \\ p & q & r \end{vmatrix} = \iint_S \frac{1}{\sqrt{3}} \left(\frac{\partial R}{\partial y} - \frac{\partial u}{\partial z} \right) - \frac{1}{\sqrt{3}} \left(\frac{\partial R}{\partial x} - \frac{\partial p}{\partial z} \right) + \frac{1}{\sqrt{3}} \left(\frac{\partial q}{\partial x} - \frac{\partial p}{\partial y} \right) ds$$

$$= \frac{1}{\sqrt{3}} \iint_S (-2y - 2z - 2x - 2z - 2x - 2y) ds$$

$$= -\frac{4}{\sqrt{3}} \iint_S (x + y + z) ds$$

$$= -\frac{6}{\sqrt{3}} \iint_S R ds$$

$$= -2\sqrt{3} \cdot \frac{3\sqrt{3}}{4} \cdot R^3 = -\frac{9}{2} R^3$$

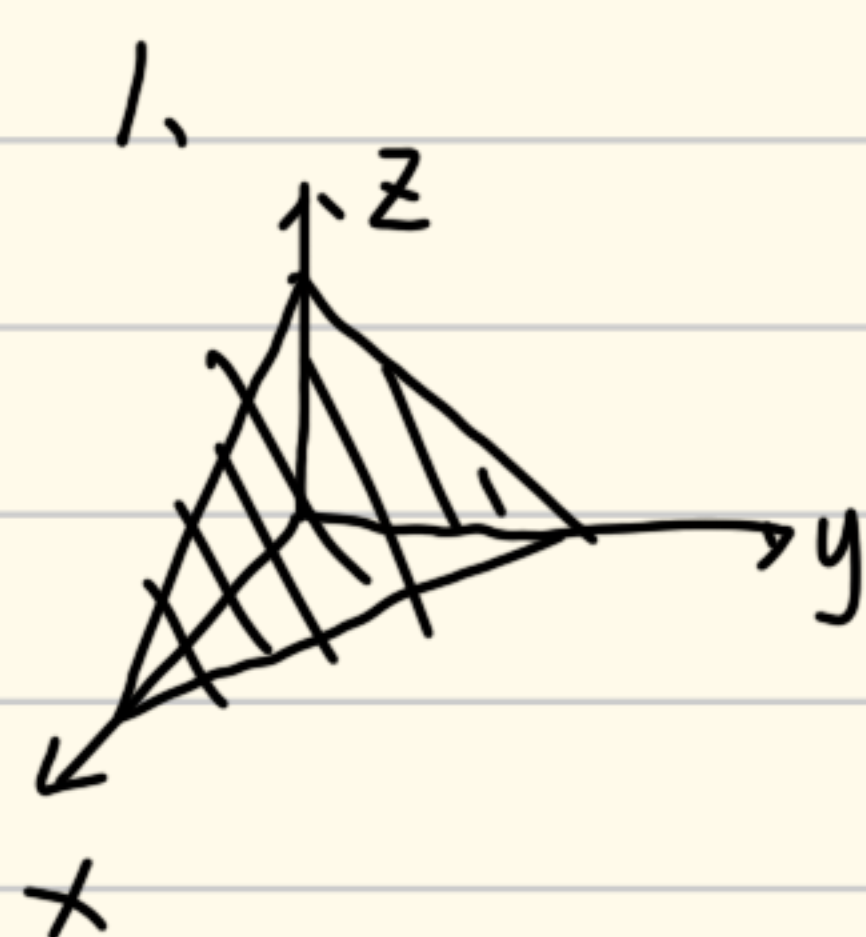


$$\frac{\sqrt{3}}{4} \cdot \frac{1}{2} R^2 = \frac{\sqrt{3}}{8} R^2 \quad 6 = \frac{3\sqrt{3}}{4} R^2$$

求下列第二型曲面积分：

1. 求 $\iint_S yz dy dz + xz dz dx + xy dx dy$ 其中 S 为平面 $x=0, y=0, z=0$ 及 $x+y+z=a (a>0)$ 所围四面体的表面外侧。

2. 求 $\iint_S xz dy dz + xy dz dx + yz dx dy$, 其中 S 是圆柱面 $x^2+y^2=R^2$ 与平面 $x=0, y=0, z=0$ 及 $z=h (h>0)$ 所围的在第一卦限中的立体的表面外侧。



$$\textcircled{1} x=0, \iint_S yz dy dz = \int_0^a dy \int_0^{a-y} yz dz = \frac{1}{2} \int_0^a (a-y)^2 y dy = \frac{1}{2} \int_0^a (y^3 - 2ay^2 + a^2y) dy$$

$$= \frac{1}{2} \cdot \left(\frac{1}{4}a^4 - \frac{2}{3}a^4 + \frac{1}{2}a^4 \right) = \frac{1}{2} \left(\frac{3}{4} - \frac{2}{3} \right) = \frac{1}{24}a^4$$

由于为内侧, 原式 $= -\frac{1}{24}a^4$

$$\textcircled{2} y=0, z=0, \text{ 由轮换对称性, 原式 } = -\frac{1}{24}a^4$$

$$\textcircled{3} x+y+z=a, \quad z=a-x-y$$

$$f_x = f_y = -1$$

$$\iint_S (-a-x-y) \cdot y \cdot (-a-x-y) \cdot x + xy \cdot (-a-x-y) dx dy = \iint_S (-ay + xy + y^2 - ax + x^2 + xy + xy) dx dy$$

$$= \iint_S (x^2 + y^2 + 2xy - ax - ay) dx dy$$

$$= \int_0^a dx \int_0^{a-x} (x^2 + y^2 + 2xy - ax - ay) dy$$

$$= \int_0^a \left[(a-x)x^2 + \frac{1}{3}(a-x)^3 + \frac{3}{2}x(a-x)^2 - ax(a-x) - \frac{1}{2}a(a-x)^2 \right] dx$$

$$= \int_0^a \left(\cancel{ax^2} - \cancel{x^3} + \frac{1}{3}a^3 - \frac{a^2x}{3} + \frac{3}{2}ax^2 - \frac{3}{2}ax^2 + \frac{3}{2}x^3 - \cancel{ax^2} + \cancel{ax^2} - \frac{1}{2}a^3 + \cancel{ax^2} - \frac{1}{2}ax^2 \right) dx$$

$$= \int_0^a \left(\frac{1}{6}x^3 + \frac{1}{2}a^2x - \frac{1}{6}a^3 \right) dx$$

$$= \frac{1}{24}a^4 + \frac{1}{4}a^4 - \frac{1}{24}a^4 = \frac{1}{4}a^4$$

则原式 $= 0$

$$\iint_S (a-x-y) \cdot y + (a-x-y) \cdot x + xy$$

$$= \iint_S a(x+y) - xy - x^2 - y^2 dx dy$$

$$= \int_0^a dx \int_0^{a-x} ax + ay - xy - x^2 - y^2 dy$$

$$= \int_0^a ax(a-x) + \frac{1}{2}a(a-x)^2 - \frac{1}{2}(a-x)^2 \cdot x - x^2(a-x) - \frac{1}{3}(a-x)^3 dx$$

$$= \int_0^a \cancel{ax^2} - \cancel{ax^2} + \frac{1}{2}a^3 - \frac{a^2x}{2} + \frac{1}{2}ax^2 - \frac{1}{2}a^2x + \cancel{ax^2} - \frac{1}{2}x^3 - \cancel{ax^2} + \cancel{ax^2} - \frac{1}{3}a^3 + \cancel{ax^2} - \frac{1}{3}ax^2 dx$$

$$= \int_0^a \left(\frac{1}{6}a^3 + \frac{5}{6}x^3 - \frac{3}{2}ax^2 + \frac{1}{2}a^2x \right) dx$$

$$= \frac{1}{6}a^4 + \frac{5}{24}a^4 - \frac{1}{2}a^4 + \frac{1}{4}a^4$$

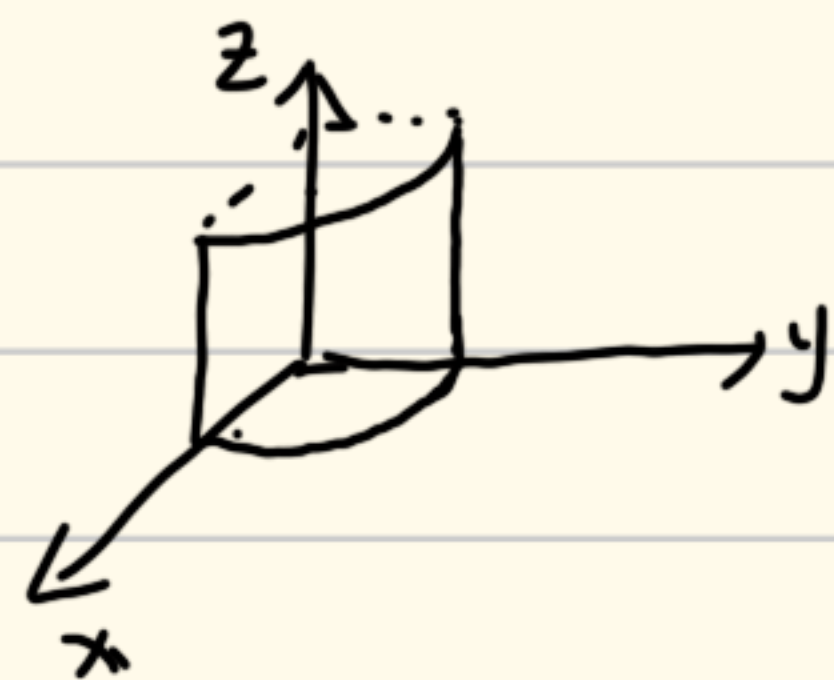
$$= \frac{4+5+6}{24} - \frac{12}{24} = \frac{1}{8}a^4$$

求下列第二型曲面积分：

1. 求 $\iint_S yz dy dz + xz dz dx + xy dx dy$ 其中 S 为平面 $x=0, y=0, z=0$ 及 $x+y+z=a (a>0)$ 所围四面体的表面外侧.

2. 求 $\iint_S xz dy dz + xy dz dx + yz dx dy$, 其中 S 是圆柱面 $x^2+y^2=R^2$ 与平面 $x=0, y=0, z=0$ 及 $z=h (h>0)$ 所围在第一卦限中的立体的表面外侧.

2.



$$1. z=0, \quad dx=dy=0, \quad \iint_S yz \, dx \, dy = 0$$

$$2. z=h, \quad dx=dy=0, \quad h \iint_S y \, dx \, dy = h \cdot \int_0^{\frac{\pi}{2}} d\theta \cdot \int_0^R r^2 \sin\theta \, dr = \frac{1}{3} R^3 h$$

$$3. y=0, \quad dx=dz=0, \quad \iint_S xz \, dy \, dz = 0$$

$$4. x=0, \quad dy=dz=0, \quad \iint_S yz \, dx \, dz = 0$$

$$5. x^2+y^2=R^2, \quad y=\sqrt{R^2-x^2}, \quad dz=0, \quad dx=\frac{-x}{\sqrt{R^2-x^2}} \, dx \quad \text{取正}$$

$$\iint_S xz \cdot \frac{x}{\sqrt{R^2-x^2}} + x \sqrt{R^2-x^2} \, dx \, dz$$

$$= \int_0^R dx \int_0^h xz \frac{x}{\sqrt{R^2-x^2}} + x \sqrt{R^2-x^2} \, dz$$

$$= \frac{1}{2} h^2 \int_0^R \frac{x^2}{\sqrt{R^2-x^2}} \, dx + h \int_0^R x \sqrt{R^2-x^2} \, dx$$

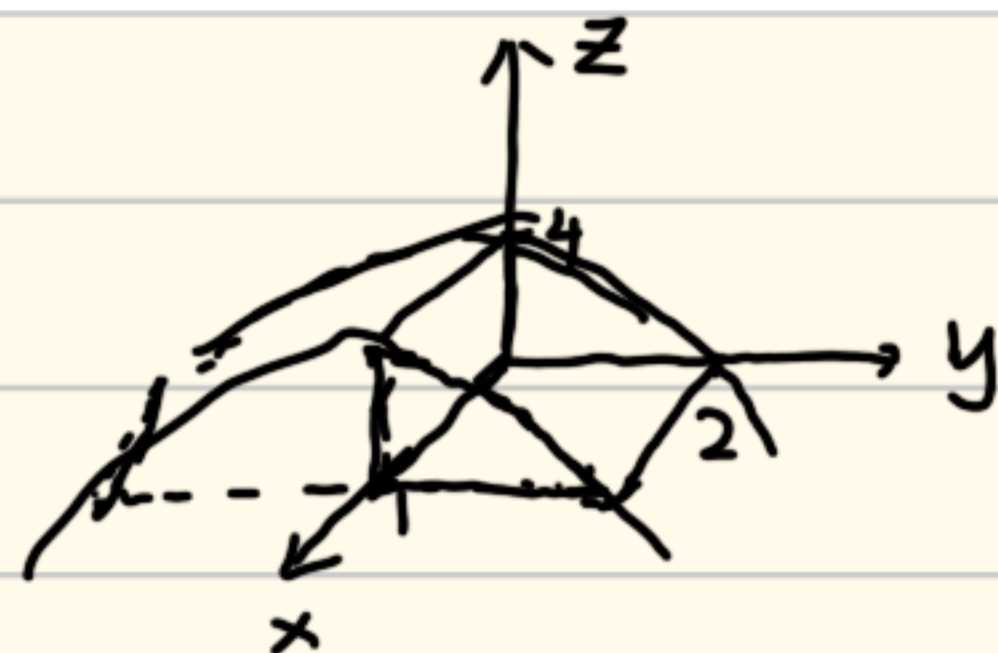
$$= \frac{1}{2} h^2 \int_0^{\frac{\pi}{2}} R^2 \sin^2\theta \, d\theta + h \int_0^{\frac{\pi}{2}} R^3 \sin\theta \cos^2\theta \, d\theta$$

$$= \frac{1}{2} h^2 R^2 \cdot \frac{1}{4} \pi + \frac{1}{3} R^3 h$$

$$= \frac{1}{8} \pi h^2 R^2 + \frac{1}{3} R^3 h$$

$$\therefore \text{原式} = \frac{1}{8} \pi h^2 R^2 + \frac{2}{3} R^3 h$$

4. 求 $\iint_S z^2 dydz + x dzdx - 3z dx dy$, 其中 S 为由曲面 $z = 4 - y^2$, 平面 $x = 0, x = 1, z = 0$, 外侧



$$S_1: y=0, S_2: z=0, S_3: x=1, S_4: x=0$$

$$S_5: z = 4 - y^2$$

$$S_1: f_x = f_z = 0, \iint_{S_1} x dz dx = \int_0^1 dx \int_0^4 x dz = 2, \text{负向}, -2.$$

$$S_2: f_x = f_y = 0, \iint_{S_2} -3z dx dy = 0$$

$$S_3: f_y = f_z = 0, \text{正向}, \iint_S z^2 dy dz$$

$$S_4: f_y = f_z = 0, \text{负向}$$

$$S_5: f_x = 0, f_y = -2y \text{ 正向}$$

$$\begin{aligned} \iint_{S_5} (2yx - 12 + 3y^2) dx dy &= \int_0^1 dx \int_{-2}^2 (2xy - 12 + 3y^2) dy = 2 \int_0^1 dx \int_0^2 (3y^2 - 12) dy \\ &= 2 \int_0^1 (8 - 24) dx \\ &= -32 \end{aligned}$$

7. 求 $\iint_S (x+a) dydz + (y+b) dzdx + (z+c) dx dy$, 其中 S 为球面 $x^2 + y^2 + z^2 = R^2$ 的外侧, a, b, c 为常数.

8. 求 $\iint_S x dydz + y dzdx + z dx dy$, 其中 S 为椭球面 $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ 的外侧.

9. 求 $\iint_S -2 dydz + 2y dzdx + e^z \sin(x+2y) dx dy$, 其中 S 是曲面 $y = e^x, 1 \leq y \leq 2, 0 \leq z \leq 2$ 的前侧.

$$7. S_1: z = \sqrt{R^2 - x^2 - y^2}, \text{正向}, S_2: z = -\sqrt{R^2 - x^2 - y^2}, \text{负向}$$

$$f_x = \frac{-x}{\sqrt{R^2 - x^2 - y^2}}, f_y = \frac{-y}{z}$$

$$f_x = \frac{-x}{z}, f_y = \frac{-y}{z}$$

两个 z 的值不一样

$$S_1: \iint_{S_1} \left(\frac{x}{z} (x+a) + \frac{y}{z} (y+b) + z+c \right) dx dy, S_2: \iint_{S_2} \left(\frac{x}{z} (x+a) + \frac{y}{z} (y+b) + z-c \right) dx dy$$

$$\text{则} = \iint_S \left(\frac{2x}{z} (x+a) + \frac{2y}{z} (y+b) \right) dx dy = 2 \iint_S \left(\frac{x}{z} (x+a) + \frac{y}{z} (y+b) + z \right) dx dy$$

$$S: x^2 + y^2 = R^2$$

由对称性,

$$\begin{aligned} \text{原式} &= 2 \iint_S \frac{x^2+y^2}{z} + z \, dx \, dy = 2 \cdot \iint_S \frac{1}{z} \cdot R^2 \, dx \, dy = 2 \iint_S \frac{R^2}{\sqrt{R^2-x^2-y^2}} \, dx \, dy \\ &= 2 \int_0^{2\pi} d\theta \cdot \int_0^R \frac{r^3}{\sqrt{R^2-r^2}} \cdot dr, \quad r = R \sin \varphi \\ &= 4\pi \cdot \int_0^{\frac{\pi}{2}} \frac{R^3 \sin^3 \varphi}{R \cos \varphi} \cdot R \cos \varphi \, d\varphi \\ &= 4\pi \cdot R^3 \cdot \frac{2}{3} = \frac{8}{3} \pi R^3 \end{aligned}$$

$$x = r \cos \theta, \quad y = r \sin \theta$$

$$2R^2 \int_0^{2\pi} d\theta \int_0^R \frac{r}{\sqrt{R^2-r^2}} \, dr$$

$$2R^2 \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} \frac{R \sin \varphi}{R \cos \varphi} \cdot R \cos \varphi \, d\varphi$$

$$= 4\pi R^3$$

7. 求 $\iint_S (x+a) \, dy \, dz + (y+b) \, dz \, dx + (z+c) \, dx \, dy$, 其中 S 为球面 $x^2 + y^2 + z^2 = R^2$ 的外侧, a, b, c 为常数.

8. 求 $\iint_S x \, dy \, dz + y \, dz \, dx + z \, dx \, dy$, 其中 S 为椭球面 $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ 的外侧.

9. 求 $\iint_S -2 \, dy \, dz + 2y \, dz \, dx + e^x \sin(x+2y) \, dx \, dy$, 其中 S 是曲面 $y = e^x, 1 \leq y \leq 2, 0 \leq z \leq 2$ 的前侧.

$$8. S_1: z = c \sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}} \quad S_2: z = -c \sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}}$$

$$\frac{x}{a^2} + \frac{z z_x}{c^2} = 0, \quad z_x = -\frac{c^2}{a^2} \frac{x}{z}, \quad z_y = -\frac{c^2}{b^2} \frac{y}{z}$$

$$\text{原式} = \iint_{S_1} \left(\frac{c^2}{a^2} \frac{x^2}{z} + \frac{c^2}{b^2} \frac{y^2}{z} + z \right) \, dx \, dy + \left(\iint_{S_2} \frac{c^2}{a^2} \frac{x^2}{z} + \frac{c^2}{b^2} \frac{y^2}{z} + z \right) \, dx \, dy$$

$$= 2 \cdot \iint_S \left(\frac{c^2}{a^2} \frac{x^2}{z} + \frac{c^2}{b^2} \frac{y^2}{z} + z \right) \, dx \, dy$$

$$= 2 \cdot \frac{c^2}{z} \cdot \iint_S \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \right) \cdot \frac{1}{z} \, dx \, dy$$

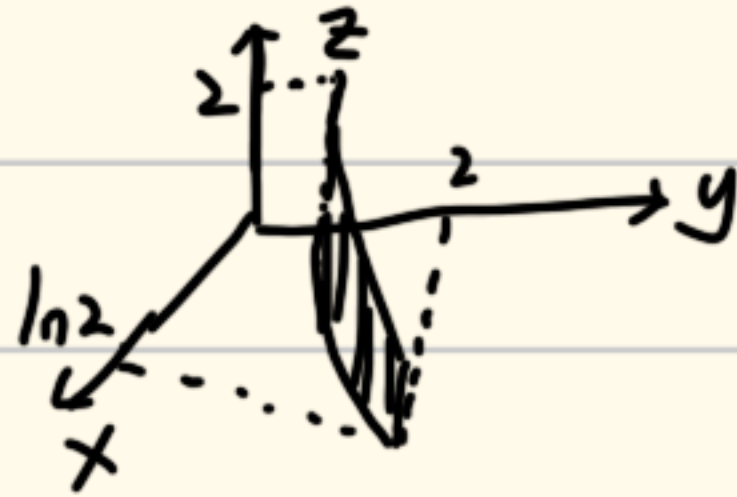
$$= 2 \iint_S \frac{c^2}{z} \cdot \frac{1}{z} \, dx \, dy$$

$$= 2c \cdot \iint_S \frac{1}{\sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}}} \, dx \, dy$$

$$= 2c \cdot \iint_S \frac{abr}{\sqrt{1-r^2}} \cdot dr \, d\theta$$

$$= 4\pi cab \int_0^1 \frac{r}{\sqrt{1-r^2}} \, dr = 4\pi abc \cdot \int_0^{\frac{\pi}{2}} \frac{\sin \varphi}{\cos \varphi} \cdot \cos \varphi \, d\varphi = 4\pi abc$$

7. 求 $\iint_S (x+a)dydz + (y+b)dzdx + (z+c)dx dy$, 其中 S 为球面 $x^2 + y^2 + z^2 = R^2$ 的外侧, a, b, c 为常数.
8. 求 $\iint_S x dy dz + y dz dx + z dx dy$, 其中 S 为椭球面 $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ 的外侧.
9. 求 $\iint_S -2dydz + 2ydzdx + e^x \sin(x+2y)dx dy$, 其中 S 是曲面 $y = e^x, 1 \leq y \leq 2, 0 \leq z \leq 2$ 的前侧.



~~前侧取正.~~ $y=e^x, f_x=e^x, f_z=0$

$$\begin{aligned} &\iint_S (2e^x + 2e^x) \cdot dx dz. \\ &= 4 \iint_S e^x dx dz \\ &= 4 \int_0^2 dz \int_0^{\ln 2} e^x dx \\ &= 8 \end{aligned}$$

1. 利用高斯公式,求第二型曲面积分
- $$\iint_S x^2 dy dz + y^2 dz dx + z^2 dx dy,$$
- 其中 S 分别为
- (1) 圆柱面 $x^2 + y^2 = a^2 (0 \leq z \leq h)$ 的外侧;
- (2) 圆锥体 $\frac{x^2}{a^2} + \frac{y^2}{a^2} - \frac{z^2}{b^2} \leq 0 (0 \leq z \leq b)$ 的全表面的外侧.
- 利用高斯公式,计算下列曲面积分:
2. 求 $\oiint_{S^+} xz^2 dy dz + (x^2 y - z^3) dz dx + (2xy + y^2 z) dx dy$, 其中 S^+ 是 $z = \sqrt{a^2 - x^2 - y^2}$ 和 $z = 0$ 所围的半球区域的全表面的外侧.
3. 求 $\oiint_{S^+} x^2 dy dz + y^2 dz dx + z^2 dx dy$, 其中 S^+ 为平面 $x = 0, y = 0, z = 0, x = a, y = a, z = a$ 所围立体之表面外侧.
4. 求 $\oiint_{S^+} x^3 dy dz + y^3 dz dx + z^3 dx dy$, 其中 S^+ 为球面 $x^2 + y^2 + z^2 = a^2$ 的外侧.

$$\begin{aligned} 2. \iiint_{S^+} (z^2 + x^2 + y^2) dV. \\ = \frac{2}{3} \pi a^3. \end{aligned}$$

$$\begin{aligned} 3. \iiint_{V.} (2x + 2y + 2z) dV \\ = 2 \int_0^a dx \int_0^a dy \int_0^a (x+y+z). dz \\ = 3a^3 \end{aligned}$$

$$\begin{aligned} 4. \quad 3 \iiint_V x^2 + y^2 + z^2 dV &= 3 \iiint (r^2 \sin^2 \varphi \cos^2 \theta + r^2 \sin^2 \varphi \sin^2 \theta + r^2 \cos^2 \varphi) r^2 \sin^2 \varphi d\varphi d\theta dr \\ &= 3 \iiint_V r^4 \sin \varphi \cdot d\varphi d\theta dr. \\ &= 6 \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} d\varphi \int_0^a r^4 (\sin^2 \varphi \cos^2 \theta + \sin^2 \varphi \sin^2 \theta + \cos^2 \varphi) \cdot \sin^2 \varphi dr. \\ &= \frac{6}{5} a^5 \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} \sin^4 \varphi \cos^2 \theta + \sin^4 \varphi \sin^2 \theta + \sin^2 \varphi \cos^2 \varphi d\varphi \\ &= \frac{6}{5} a^5 \left(\frac{3}{16} \pi + \frac{\pi}{16} \right) \cdot 2\pi. \\ &= \frac{3}{5} a^5 \pi^2 \end{aligned}$$

7. 求 $\oiint_{S^+} (xy^2 + y + z) dy dz + (yz^2 + xz) dz dx + (zx^2 + 5x^2 y^2) dx dy$, 其中 S^+ 为

椭球面 $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ 的外侧.

8. 求第二型曲面积分 $\iint_{S^+} \mathbf{F} \cdot \mathbf{n} dS$, 其中 $\mathbf{F} = \sqrt{x^2 + y^2 + z^2} (x, y, z)$, S^+ 是区域:

$\{(x, y, z) \mid 1 \leq x^2 + y^2 + z^2 \leq 2\}$ 的边界面, 指向外侧.

$$\begin{aligned}
 7. \quad & \iiint_V (y^2 + z^2 + x^2) dV, \quad x = a r \sin \varphi \cos \theta, \quad y = b r \sin \varphi \sin \theta, \quad z = c r \cos \varphi \\
 & \iiint_V (a^2 r^2 \sin^2 \varphi \cos^2 \theta + b^2 r^2 \sin^2 \varphi \sin^2 \theta + c^2 r^2 \cos^2 \varphi) abc r^2 \sin \varphi dr d\varphi d\theta \\
 & = 2abc \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} d\varphi \int_0^1 r^4 \sin \varphi (a^2 \sin^2 \varphi \cos^2 \theta + b^2 \sin^2 \varphi \sin^2 \theta + c^2 \cos^2 \varphi) dr \\
 & = \frac{2}{5} abc \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} a^2 \sin^3 \varphi \cos^2 \theta + b^2 \sin^3 \varphi \sin^2 \theta + c^2 \sin \varphi \cos^2 \varphi d\varphi \\
 & = \frac{2}{5} abc \int_0^{2\pi} \left(\frac{2}{3} a^2 \cos^2 \theta + \frac{2}{3} b^2 \sin^2 \theta + \frac{1}{3} c^2 \right) d\theta \\
 & = \frac{2}{5} abc \left(\frac{2}{3} a^2 \pi + \frac{2}{3} b^2 \pi + \frac{2}{3} c^2 \pi \right) \\
 & = \frac{4}{15} (a^2 + b^2 + c^2) abc \pi
 \end{aligned}$$

$$\begin{aligned}
 8. \quad & \iiint_V \sqrt{x^2 + y^2 + z^2} dV \\
 & 2 \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} d\varphi \int_1^{\sqrt{2}} r^2 \sin \varphi \cdot \frac{r \sin \varphi \cos \theta + r \sin \varphi \sin \theta + r \cos \varphi}{r} dr \\
 & \quad r^2 (\sin^2 \varphi \cos \theta + \sin^2 \varphi \sin \theta + \sin \varphi \cos \varphi) dr \\
 & 2 \cdot \frac{2}{3} \cdot 2\pi \cdot \frac{1}{2} =
 \end{aligned}$$

12. 求 $\oint_{L^+} y^2 dx + xy dy + xz dz$, 其中 L 是圆柱面 $x^2 + y^2 = 2y$ 与平面 $y = z$ 的交线, 逆时针方向.

13. 求 $\oint_{L^+} (y^2 + z^2 - x^2) dx + (x^2 + z^2 - y^2) dy + (x^2 + y^2 - z^2) dz$, 其中 L 为球面 $z = \sqrt{2Rx - x^2 - y^2}$ 与柱面 $x^2 + y^2 = 2rx$ ($0 < r < R, z \geq 0$) 的交线, L^+ 与球面上侧成右手系.

14. 求 $\oint_{L^+} (y^2 - z^2 + x^2) dx + (z^2 - x^2 + y^2) dy + (x^2 - y^2 + z^2) dz$, 其中 L 是平面 $x + y + z = \frac{3}{2}R$ 与立方体 $\{(x, y, z) | 0 \leq x \leq R, 0 \leq y \leq R, 0 \leq z \leq R\}$ 的交线, 从 x 轴正向看去, L^+ 沿逆时针方向 (见图 8.50).

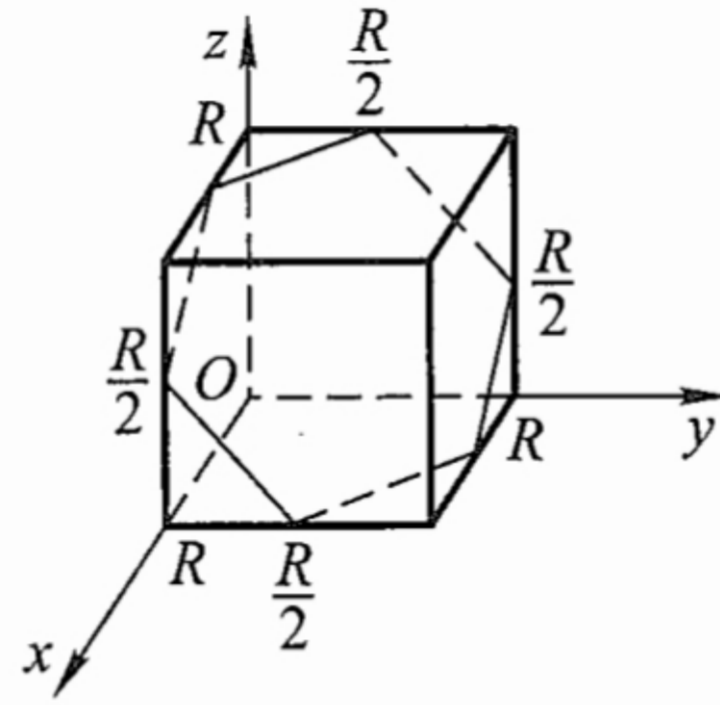
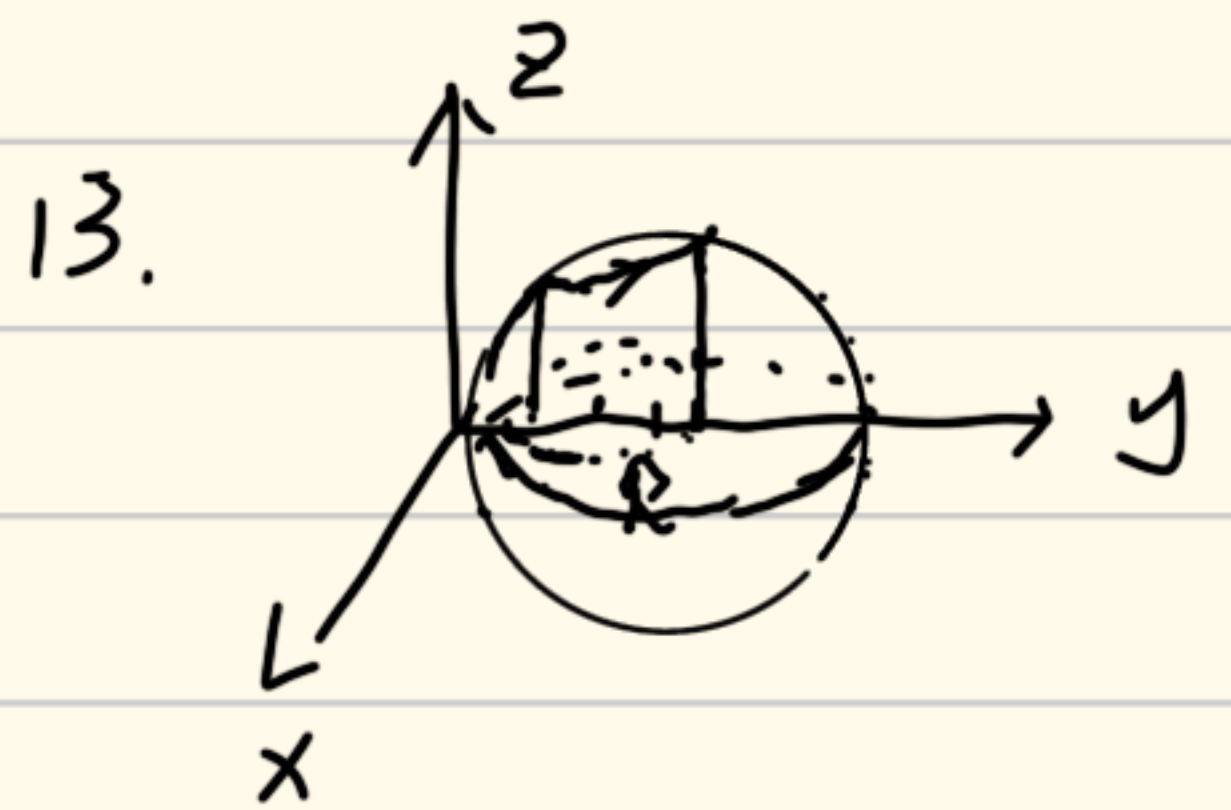
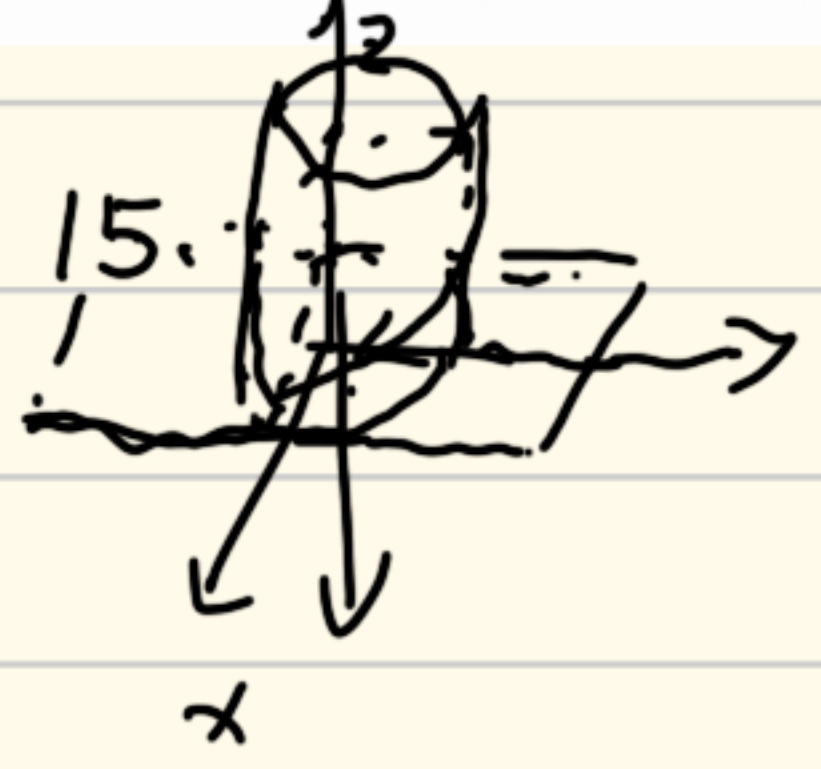


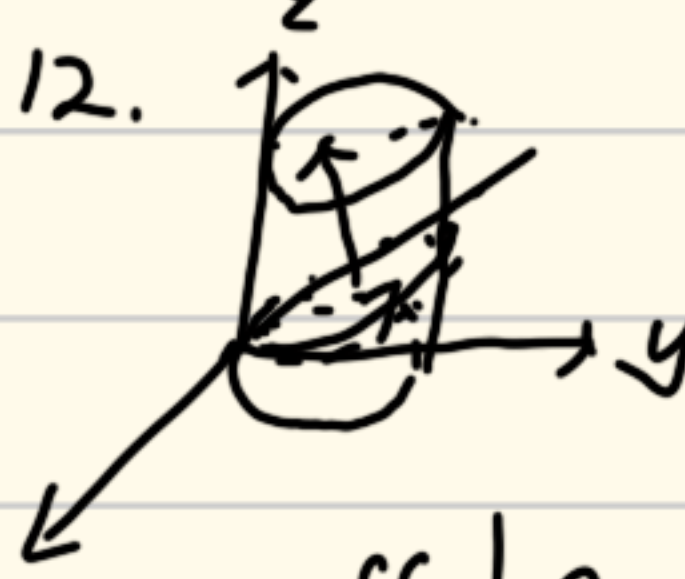
图 8.50

15. 求 $\oint_{L^+} (y - z + x) dx + (z - x + y) dy + (x - y + z) dz$, 其中 L^+ 为椭圆周: $x^2 + y^2 = 1, x + z = 1$, 沿顺时针方向.

16. 求 $\oint_{L^+} xz dx + xy dy + 3xz dz$, 其中 L 为平面 $2x + y + z = 2$ 在第一卦限部分的边界, 从 x 轴正向看去, 沿逆时针方向.



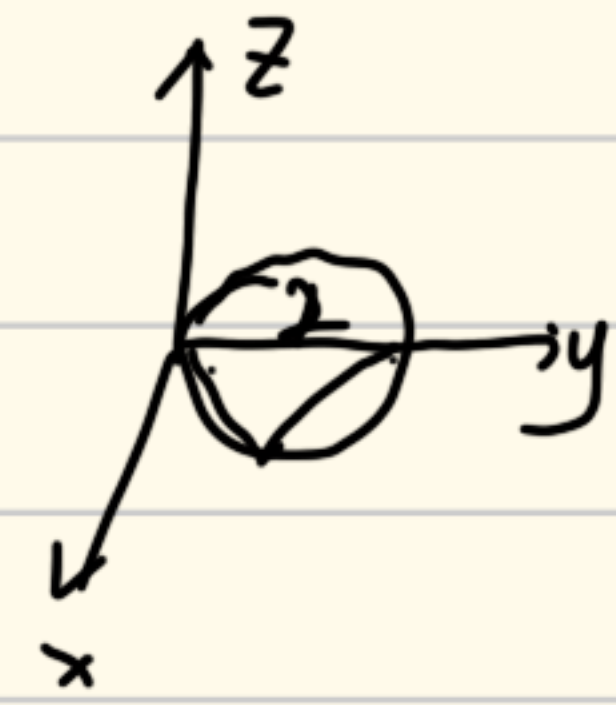
13.



法向量 $(0, 1, 1)$

$$y = z \quad y - z = 0$$

$$\begin{aligned} \iint_S \begin{vmatrix} 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 & xy & xz \end{vmatrix} ds &= \frac{1}{\sqrt{2}} \iint_S (z + y - 2y) ds = \frac{1}{\sqrt{2}} \iint_S (z - y) ds \\ &= \frac{1}{\sqrt{2}} \iint_S \sqrt{0+1+1} \cdot (z - y) d\theta \\ &= \iint_S -y d\theta = \int \end{aligned}$$



$$r^2 = 2r \sin \theta$$

$$r = 2 \sin \theta$$

$$\frac{1}{\sqrt{2}} (1, 0, 1)$$

$$\begin{aligned} -\frac{1}{\sqrt{2}} \iint_S \begin{vmatrix} 1 & 0 & 1 \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y-z+x & z-x+y & x-y+z \end{vmatrix} ds &= -\frac{1}{\sqrt{2}} \iint_S (-1-1-1-1) ds = \frac{4}{\sqrt{2}} \iint_S ds \\ &= \frac{4}{\sqrt{2}} \cdot 2 \cdot \sqrt{2} = 4\sqrt{2} \end{aligned}$$

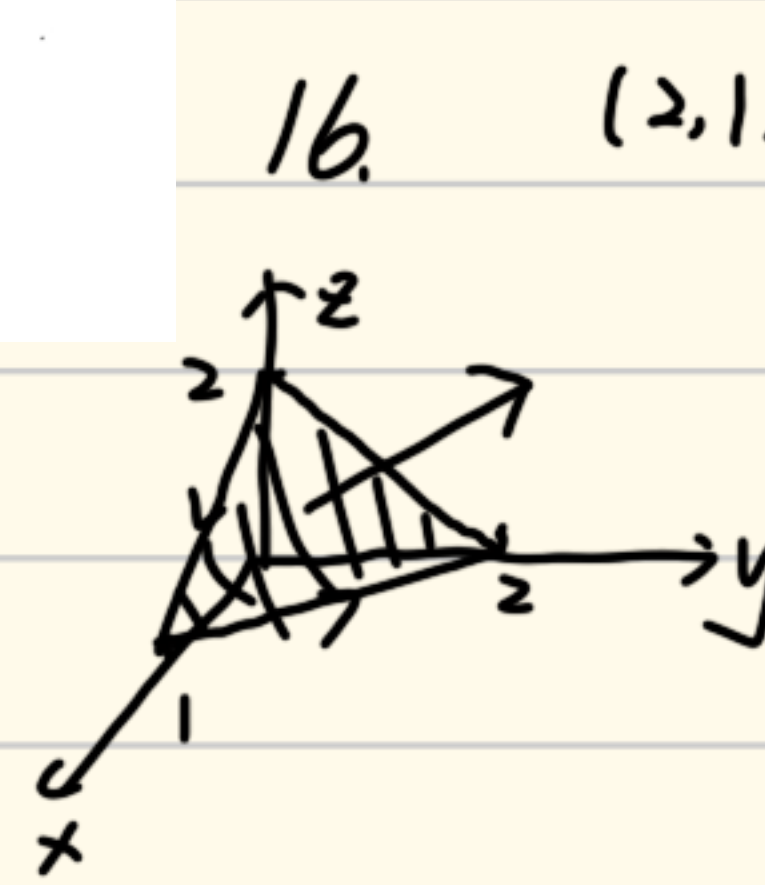
$$b = 1, a = \sqrt{2}$$

$$\begin{aligned} \iint_S \begin{vmatrix} dy dz & dx dz & dx dy \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 + z^2 - x^2 & x^2 + z^2 - y^2 & x^2 + y^2 - z^2 \end{vmatrix} &= \iint_S (2y - 2z) dy dz - (2x - 2z) dx dz + (2x - 2y) dx dy \\ z &= \sqrt{2Rx - x^2 - y^2} \\ z_x &= \frac{2R - 2x}{2\sqrt{2R - 2x}} = \frac{R - x}{\sqrt{2R - 2x}} \end{aligned}$$

$$\begin{aligned} &= \iint_S (2y - 2z) dy dz - (2x - 2z) dx dz + (2x - 2y) dx dy \\ &= 2 \iint_S \left(\frac{R-x}{2} (z-y) + \frac{y}{2} (z-x) + x-y \right) d\theta \\ &= 2 \iint_S R - x + \frac{xy - Ry}{2} + y - \frac{xy}{2} + x - y d\theta \\ &= 2 \iint_S R - \frac{R}{2} y d\theta \end{aligned}$$

$$\begin{aligned}
 x^2 + y^2 &= 2Rx \\
 r &= 2R \cos \theta \\
 z &= \sqrt{2R \cdot r \cos \theta - r^2} \\
 &= 22Rr^2 - 2R \iint \frac{y}{z} d\theta
 \end{aligned}$$

16. 求 $\oint_{L^+} xzdx + xydy + 3xzdz$, 其中 L 为平面 $2x + y + z = 2$ 在第一卦限部分的边界, 从 x 轴正向看去, 沿逆时针方向.



$$\begin{aligned}
 & \frac{1}{\sqrt{6}} \iint_S \begin{vmatrix} 2 & 1 & 1 \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xz & xy & 3xz \end{vmatrix} dS = \frac{1}{\sqrt{6}} \iint_S (x - 3z + y) dS \\
 & \quad \quad \quad z = 2 - y - 2x \\
 & = \frac{1}{\sqrt{6}} \cdot \sqrt{4+1+1} \iint_S (x - 6 + 3y + 6x + y) d\theta \\
 & = \iint_S (7x + 4y - 6) d\theta \\
 & = \int_0^1 dx \int_0^{2-2x} (7x + 4y - 6) dy \\
 & = \int_0^1 8 \cdot (1-x)^2 + 7x(2-2x) - 6(2-2x) \cdot d|x \\
 & = \int_0^1 8 - \underline{16x} + 8x^2 - 12 + \underline{12x} \\
 & \quad \quad \quad + \underline{14x} - 14x^2 \cdot \\
 & = \int_0^1 (-6x^2 + 10x - 4) dx \\
 & = -2 + 5 - 4 \\
 & = -1
 \end{aligned}$$